

01 PROBLEM ANALYSIS

Problem Analysis

Who TRINETRA serves, the precise disaster-management problems it addresses, and the product response.

Problem statement context

SIH Problem Statement ID: 26206

Theme: Disaster Management

Category: Software

Organization: AICTE / MIC–Student Innovation

TRINETRA is planned as a unified disaster-management platform that turns scattered observations and operational data into clear, location-aware actions for citizens and authorities.

Primary users

- Citizens who need timely alerts, local risk context, safe zones, and emergency resources.
- Authorities who need a consolidated view of disasters, citizen reports, affected areas, resources, and alerts.
- Responders who need location context and prioritized, traceable information.

Specific problems to solve

Problem	Operational effect	TRINETRA response
Fragmented information	No single current picture across incidents, alerts, maps, and resources.	Centralized disaster dashboard.
Lack of localized information	Users cannot easily understand what applies to their area.	Location-based map, affected areas, risk, and resources.
Delayed alerts	Citizens may not receive timely, actionable warnings.	Authority-controlled alert publishing and citizen alert view.
Difficulty finding safe locations	People may not know where to go during an emergency.	Safe-zone visualization and resource discovery.
Emergency resources are hard to find	Hospitals, shelters, and relief centers may be difficult to locate.	Map and list of verified emergency resources.
Citizen information does not reach authorities efficiently	Early local observations may remain unstructured or unseen.	Structured citizen incident reporting.
Limited situational awareness	Authorities need a combined view of incidents, reports, risk, and geography.	Authority dashboard plus map, reports, and risk layer.

Problem-analysis rule

Do not claim impact or cite statistics unless the team has a verified source. Separate validated facts, assumptions, and design hypotheses.